Activity 1.8 – Synchronizer, Edge Detector and Debouncer

1. Design Approach:

Designed three input-conditioning circuits using Verilog HDL: a two-stage synchronizer, a rising-edge detector and a counter-based debouncer. The synchronizer handles asynchronous inputs, the edge detector generates a one-clock pulse, and the debouncer removes unwanted input fluctuations.

2. Verification Steps:

- Created a separate testbench.
- Tested synchronizer operation and clock-cycle delay.
- Verified rising-edge pulse generation.
- Checked pulse deactivation.
- Tested debouncer operation with a stable input.
- Simulated using ModelSim.

3. Results:

The testbench checks synchronizer output, edge detection pulse and debounced output. PASS/FAIL messages are displayed.

4. Issues Faced:

Synchronizer stages must be connected in sequence. The edge detector must generate only one pulse for each rising edge. The debounce counter must be configured correctly.

5. Conclusion:

Successfully designed and verified a synchronizer, rising-edge detector and counter-based debouncer using Verilog HDL.